7. *In* ***Section E****, Question numbers* ***36-38*** *are case study based questions, carrying 04 marks each with sub parts of the values of* ***1, 1 and 2 marks each respectively****.*

8. ***There is no overall choice****. However an internal choice in 2 questions of Section B, 2 questions of Section C and 2 questions of Section D has been provided in all the 2 marks questions of Section E.*

9. *Draw neat and clean figures whereeever required.*

   Take $\pi = \frac{22}{7}$ *wherever required if not stated.*

10. ***Use of calculators is not allowed.***

Section A

(Section A consists of 20 questions of 1 mark each.)

1. The greatest number which divides both 30 and 80, leaving remainders 2 and 3 respectively is

   a) 10   b) 7

   c) 308   d) 11

2. For what value of p, the following pair of linear equations has infinitely many solutions?

$$10x + 5y - (p - 5) = 0$$

$$20x + 10y - p = 0$$

a) 10 b) –10

c) 5 d) –5

3. If $\sin A = \frac{\sqrt{3}}{2}$ and A is an acute angle, then find the value of $\frac{\tan A - \cot A}{\sqrt{3} + \operatorname{cosec} A}$.

a) $-\frac{2}{5}$ b) $\frac{2}{5}$

c) $\frac{2}{3 + 2\sqrt{3}}$ d) –2

4. If (1, –3), (–2, –1) and (–2, 2) are the three vertices of a parallelogram taken in that order, then the fourth vertex is

a) (–1, –2) b) (1, 2)

c) (–1, 2) d) (1, 0)

5. The sum of exponents of prime factors in the prime-factorisation of 675 is :

a) 3 b) 4

c) 5 d) 2

6. $\sin^4\theta + \cos^4\theta = \frac{1}{2}$ then find $\sin\theta\cos\theta$,

$(0 \le \theta \le 90^\circ)$

a) $\frac{1}{8}$ b) $\frac{1}{4}$

c) 1 d) $\frac{1}{2}$

7. Find the mode of the following discrete series

X :	1	3	5	6	12	15
f :	5	7	3	8	6	5

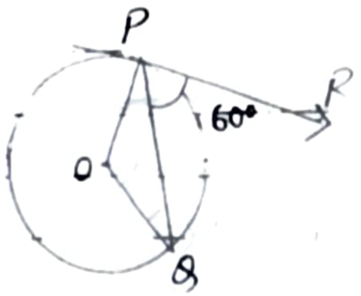
a) 3 b) 12

c) 8 d) 6

8. If $x^2 + k(2x + k - 1) + 2 = 0$ has equal roots, then K =

a) −2 b) 2

c) $\frac{1}{2}$ d) $\frac{-1}{2}$

9. 

In the figure if PR is tangent to the circle and O is the centre of the circle $\angle QPR = 60°$ then $\angle POQ = ?$

a) $110°$ b) $100°$

c) $120°$ d) $90°$

10. A card is drawn at random from a well shuffled pack of 52 playing cards. The probability of getting a red face card is

a) $\frac{3}{25}$ b) $\frac{3}{26}$

c) $\frac{3}{28}$ d) none of these

11. If α and β are the zeros of $kx^2 - 2x + 3k$ such that $\alpha + \beta = \alpha\beta$ then K=

a) $\frac{1}{3}$ b) $\frac{-1}{3}$

c) $2/3$ d) $-2/3$

12. 

O is the point of intersection of two chords AB and CD of a ~~circel~~ circle. If $\angle BOC = 80^\circ$ and OA = OD then ΔODA and ΔOBC are

a) equilateral and similar

b) isosceles and similar

c) isosceles but not similar

d) not similar

13. $3\cos^2 60^\circ + 2\cot^2 30^0 - 5\sin^2 45^\circ =$

a) $\frac{17}{4}$

b) $\frac{13}{6}$

c) 1

d) 4

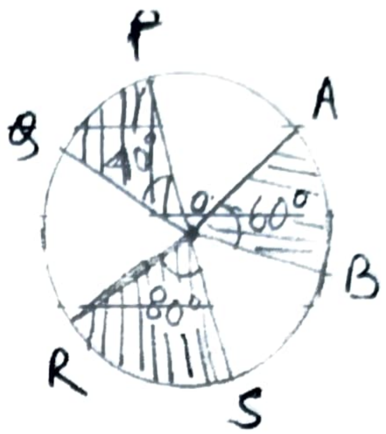
14. The median and mode of a frequency distribution are 26 and 29 respectively. Then the mean is

a) 27.5

b) 24.5

c) 28.4

d) 25.8

15. 

In the given figure, the radius of the ~~cricle~~ circle with centre O is 7 cm, If $\angle POQ = 40^0$, $\angle AOB = 60^0$ $\angle ROS = 80^0$, the area of the shaded region

a) $11 cm^2$ b) $22 cm^2$

c) $49 cm^2$ d) $77 cm^2$

16. The length of the arc is one-twelth that of circuference of a circle. Then the angle subtended by the arc at the centre of the circle is

a) 30° b) 60°

c) 36° d) 45°

17. The volumes of two spheres are in the ratio 64 : 27 The ratio of their surface areas is

a) 9 : 16 b) 16 : 9

c) 3 : 4 d) 4 : 3

18. If points A(4, 3) and B(x, 5) are on a circle with centre O (2, 3), then the value of $x^2 + 5$ is

a) 9
b) 14
c) 10
d) 21

Directions : In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option :

a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

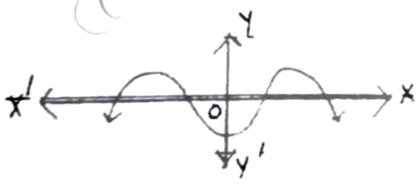
b) Both assertion (A) and reason (R) are true reason (R) is not the explanation of assertion (A).

c) Assertion (A) is true but reason (R) is false.

d) Assertion (A) is false but reason (R) is true.

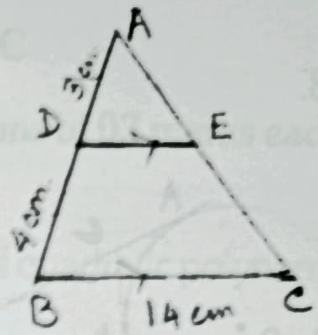
19. **Assertion (A)** : The graph $y = f(x)$ is chown in figure, for the polymomial f(x). The number of zeros of f(x) is 4.

(c)



Reason (R) : The number of zero of the polymomial f(x) is the number of points at which f(x) cuts or touches axes.

20. **Assertion (A)** : In the given figure, if DE||BC, AD = 3cm, BD = 4cm and BC = 14cm, then DE = 6cm.



Reason (R) : In ΔABC, if DE||BC then $\Delta ADE \sim \Delta ABC$.

(a

Section - B

Section B consists of 5 questions of 2 marks each.

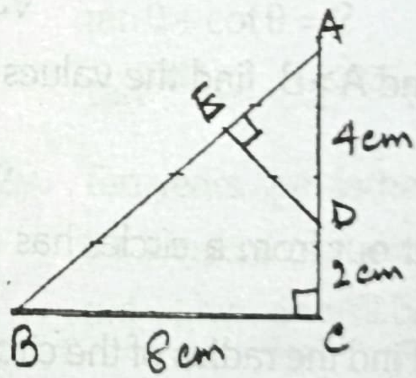
21. In an AP if $S_n = n(4n+1)$ then find the AP.

5, 13, 21, 29

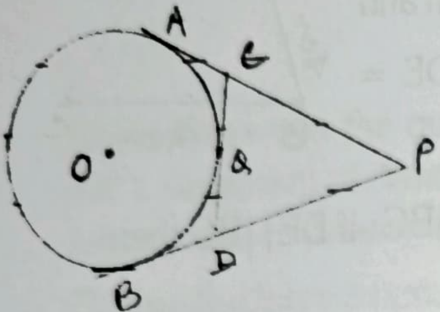
Or

Find the 6th term from the end of the AP 17, 14, 11, ... (–40).

22. In the adjacent figure, ABC is a right angled triangle in which $\angle ACB = 90^\circ$, $DE \perp AB$. Find the length of AE. = 2.4 cm



23. PA and PB are tangent to the circle from an external point P. CD is another tangent touching the circle at Q. If PA = 12cm, QC = QD = 3 cm, find PC+ PD= 18cm



24. Find the value of θ, if $\frac{1}{\sec\theta - 1} - \frac{1}{\sec\theta + 1} = \frac{2}{3}$, where $0° < \theta < 90°$

Or

If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, $0° < A+B < 90°$ and A>B, find the values of A and B. 37.5

45

25. A sector of 120^0 cut out from a circle, has an area of $9\frac{3}{7}$ sq.cm. Find the radius of the circle.

3 cm

Section - C

Section C consists of 6 questions of 03 marks each.

26. If $\frac{p}{2}$ and $\frac{q}{2}$ are the zeros of quadratic polynomial $3x^2+5x+7$, then find a quadratic polynomial whose zeros are $\frac{1}{p}$ and $\frac{1}{q}$.

27. Prove that the angle between two tangents drawn from ~~a~~ an external point to a circle is supplementary to the angle subtended by the line segments joining the points of contact at the centre.

28. Prove that, $\frac{\cot A+\operatorname{cosec} A-1}{\cot A-\operatorname{cosec} A+1}=\frac{1+\cos A}{\sin A}$

Or

$\tan\theta+\cot\theta=2$, find the value of $\tan^{2025}\theta+\cot^{2025}\theta$.

29. Ten years ago, father was twelve times as old as his son and ten years hence, he will be twice as old as his son will be. Find their present ages.

Or

son = 12
father = 34

Solve : $\dfrac{5}{x+y} - \dfrac{2}{x-y} = -1$

$$\frac{15}{x+y} + \frac{7}{x-y} = 10$$

30. A bag contains white, black and red balls only. A ball is drawn at random from the bag. The probability of getting a white ball is $\frac{3}{10}$ and that of a black ball is $\frac{2}{5}$. Find the probability of getting a red ball. →3/10 If the bag contains 20 black balls, find the total numbers of balls in the bag. →50

31. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds respectively. If they all change simultaneously at 8 am, then at what time will they again change siumtaneously? 72 mins past 8 am

Section - D

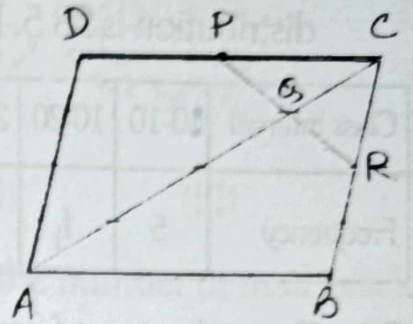
Section D consists of 4 questions of 05 marks each

32. A fast train takes 3 hours less than a slow train for a journey of 600km. If the speed of the slow train is 10km/hr less than that of the fast train find the speeds of the two trains.

fast = 50 km/hr

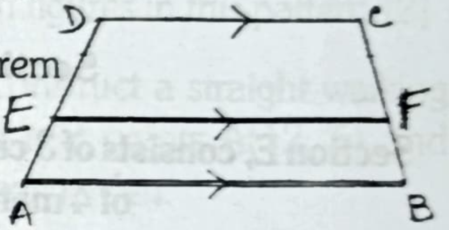
slow = 40 km/hr

33. ABCD is a parallelogram in which P is the mid point of DC and Q is a point on AC such that $CQ = \frac{1}{4} AC$. If PQ produced meets BC ~~and~~ at R, prove that R is the midpoint of BC.



Or

State Thales theorem. In trapezium ABCD, AB||DC and EF||DC.



Using the above theorem prove that

$$\frac{AE}{ED} = \frac{BF}{FC}$$

34. If the mean of the following distribution is 27. Find the value of p.

Class	0-10	10-20	20-30	30-40	40-50
Frequency	8	p	12	13	10

Or

If the median of the following frequency distribution is 28.5. Find the missing frequencies.

Class Interval	0-10	10-20	20-30	30-40	40-50	50-60	Total
Frequency	5	f_1	20	15	f_2	5	60

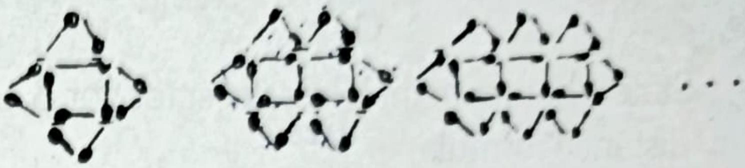
35. From the top of a vertical tower, the angles of depression of two cards in the same straigth line with the base of the tower, at an instant are found to be $45°$ and $60°$. If the cars are 100m apart and are on the same side of the tower, find the height of the tower. → $50\sqrt{3}$ m

Section - E

Section E, consists of 3 case-study based questions of 4 marks each.

36. In Mathematics, relations can be expressed in various ways. The match stick patterns are based on linear relation. Different strategies can be used to calculate the number of matchsticks used in different figures.

One such pattern is shown below. Observe the pattern and answer the following questions using Arithmetic Pregression :



(I) (II) (III) . . .

i) Write the AP for the number of matchsticks used in this pattern, [1] =12 d=7

ii) Write the 10th term of the AP. =75 [1]

iii) Which figure has 61 matchsticks? [2]

Or

Find the total number of matchsticks required to construct 15 such figures in this pattern. [2] =915

37. A school plans to construct a straight walking track between two rest points A(12, 5) and B(4, –3) in the school garden.

To install a drinking water top, the school decides to place it at a point P(x, 2) on the line joining A and B such that the tap is easily accessible from any one both the ends.

From the given information, answer the following questions.

i) In what ratio does P divide line AB? 3:5 [1]

ii) Using the ratio, find the abscissa of point P. → 8 [1]

iii) Find the distance between two end points A & B. $8\sqrt{2}$ [2]

Or

State whether point P ~~lids~~ lies closer to A or B, using distance formula. [2]

38. Kabya's mother gives an oder of double layer cake for her 16th Birthday in the shape of cylinder with base diameter 24cm and height 12cm, with another cylindrical top having diameter 16cm and height 8 cm. The baker wants to cover the cake with blue whipped cream.

From the above information answer the following questions.

i) Calculate the curved surface area of the top layer. 402.28 cm² [1]

ii) Calculate the curved surface area of the bottom layer. 905.12 cm² [1]

iii) Find the surface area that will be decorated with blue whipped cream. [2]

Or

Find the total volume of the cake. [2]

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